

IT Infrastructure Design and Deployment Plan



StarTech Business College Enterprise IT Architecture Simulation

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1. EXECUTIVE SUMMARY

This document presents the end-to-end IT infrastructure design for StarTech Business College, developed as a structured enterprise-style deployment plan. The solution defines a complete campus-wide technology environment integrating networking, server infrastructure, endpoint systems, cloud identity services, and security architecture.

The proposed design establishes a scalable and secure IT ecosystem capable of supporting academic, administrative, and operational workloads across multiple user groups, including students, teaching staff, administrative personnel, and management.

The infrastructure incorporates enterprise networking components such as managed switches, wireless access points, firewalls, and structured cabling, along with centralized server systems and endpoint devices. In addition, Microsoft 365 cloud integration provides identity management, domain services, and role-based access control across the organization.

The design also includes a detailed procurement and cost model, ensuring alignment with real-world IT budgeting and infrastructure planning practices. Overall, this solution demonstrates a complete enterprise-grade IT environment designed for scalability, security, and operational efficiency.

2. PROJECT OVERVIEW

StarTech Business College requires a fully integrated IT infrastructure to support modern educational operations across multiple departments and user roles. This project outlines the design and implementation approach for establishing a secure, scalable, and centrally managed campus IT environment.

The architecture combines on-premises infrastructure with cloud-based services to ensure high availability, centralized identity management, and efficient resource utilization. Key components include server infrastructure, endpoint systems, structured network topology, wireless connectivity, surveillance systems, and cloud identity services.

The environment is designed to simulate real-world enterprise IT operations, incorporating best practices in network segmentation, security enforcement, access control, and infrastructure standardization.

3. INFRASTRUCTURE REQUIREMENTS

The following hardware components define the required IT environment for the institution:

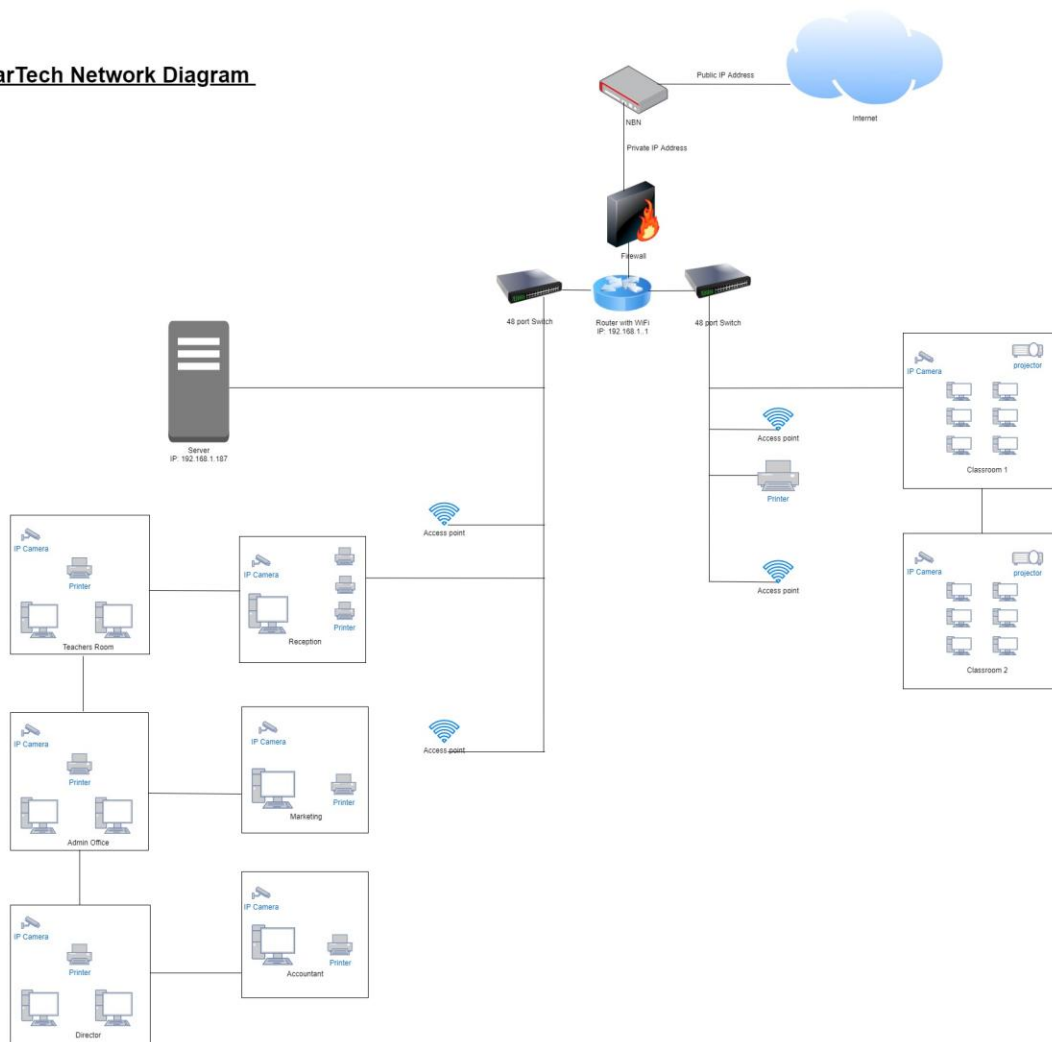
Component	Quantity
Staff Computers	8
Server Room Computer	1
Student Computers	60

Extended Monitors	2
IP Cameras	8
Switches	2
Printers	9
Projectors	2
Access Points	4
Router	1
Firewall	1
Servers	2

4. NETWORK ARCHITECTURE DESIGN

The network architecture is structured to ensure secure, segmented, and high-performance connectivity across the campus environment.

StarTech Network Diagram



5. HARDWARE & INFRASTRUCTURE SPECIFICATIONS

Item Required	Provider/Supplier/Brand/Model	Description/Comments	Estimated Cost in AUD
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Domain	startechbusiness.college	Unique domain for college's online presence	\$3.65 for 1 st year than \$17 per year
<u>Computers</u> Classrooms & Server 61* \$999.00	HP 23.8-inch All-in-One Desktop PC 24-cr0002a	Equipped with Intel® Celeron® processor, Windows 11 Home, 23.8" FHD display, 8 GB RAM, 256 GB SSD, and HP 125 USB peripherals. Ideal for classroom and server use. Link: Shopping Cart - Shop HP.com Australia	\$71,729.00
Staff Members 8 * \$1279.00	HP 23.8-inch All-in-One Desktop PC 24-cr0001a	Features 13th Generation Intel® Core™ i5 processor, Windows 11 Home, 23.8" FHD display, 16 GB RAM, 512 GB SSD, and HP 125 USB peripherals. Tailored for staff use. Link: HP 23.8 inch All-in-One Desktop PC 24-cr0001a - 23.8" (911K6PA) - Shop HP.com Australia	
Extendable Monitors 2 * \$279.00	HP M24fe 23.8 inch FHD Monitor	23.8" FHD IPS display with Edge-lit, VGA, HDMI, tilt stand, AMD FreeSync™, and low blue light mode. For Director and Receptionist. Link : You left 1 item(s) in your cart! (hp.com)	
<u>Printer (9)</u> (Total 7 departments are including students dept) Black & White Printer \$660.00 * 7) ❖ Admin office ❖ Teacher's Room	HP LaserJet Pro MFP 4101fdn Printer	A4 Black and White Laser Multifunction Printer, perfect for business use with print, scan, copy, and fax capabilities. Supports USB and Ethernet connections. Link: You left 1 item(s) in your cart! (hp.com)	\$7,713.00.
Colorful Laser Printer \$792.00) ❖ Reception	HP Color LaserJet Pro MFP 4301dw Printer	A4 Colour Laser Multifunction Printer, ideal for business with print, scan, and copy functions. Supports Wi-Fi and duplex printing. Link: HP Color LaserJet Pro MFP 4301dw Printer - (4RA80F) - Shop HP.com Australia	
Plastic ID Printer \$1999.00) PVC Cards (\$119) Evolis Zenius PVC ID cards 0.76mm Thick White 100 Pack Color Ribbons (\$183)	Evolis Zenius CLASSIC Single Sided USB ID Card Printer Bundle	For creating student and staff ID cards with full-color edge-to-edge printing. Includes software, USB connectivity, and starter pack with blank cards and color ribbon. Link: Evolis Zenius ID Card Printer Kit Officeworks	

Evolis Primacy 6 Panel Colour Ribbon Cartridge			
<u>Digital Camera</u>	Nikon Coolpix P1000 125x Zoom Digital Camera	For taking high-quality student photos for IDs. Features 4K video recording. Link: Nikon Coolpix P1000 125x Zoom Digital Camera [4K Video] - JB Hi-Fi (jbhifi.com.au)	\$1499
<u>Projectors</u> \$699.00*2	Epson EB-W06 WXGA 3LCD Projector White	WXGA resolution projectors with 3,700 lumens brightness and built-in stereo speaker. Supports HDMI, WiFi, and split screen technology. Link: Epson EB-W06 WXGA 3LCD Projector White Officeworks	
Projector Screen \$198*2	Compatible with Epson Projectors		
<u>Internet Service</u> Main ISP \$85 per month (\$85 *12 Months)	Optus (Fibre) 240 Mbps	Unlimited Data	\$1860 yearly
Backup \$70 per month (\$70 *12 Months)	Belong Mobile Broadband 400GB	5G/ 4G / 3G	
<u>CCTV System</u> (\$191.20 * 8)	Hikvision DS- 2CD2143G2-I 2.8mm Fixed Dome AcuSense Camera	The Hikvision DS-2CD2143G2-I combines 4 MP clarity, backlight compensation, and H.265+ efficiency in a robust, weather-resistant design. Link: Amazon.com.au Shopping Cart	\$1529.6
<u>Server Rack</u>	4Cabling	Reversible tempered glass door, pre-assembled, cold-rolled steel. Meets ANSI-310-D, IEC 297-2 standards. Link: 42RU 600mm Wide x 800mm Deep Server Rack 4Cabling	\$1,263.01
<u>Server</u>	HPE DL380 Gen10 Plus (\$4,194.87)	Ideal for virtualization and database management. Intel Xeon Silver 4310, 8 SFF bays, MR416i-p RAID.	\$10,193.74

	HPE ProLiant DL360 Gen10 (\$5,998.87)	Dense, efficient for applications, storage. Intel Xeon Silver, 26.4 TB storage, 16GB DDR4. Link: HPE ProLiant DL360 Gen10 Server 26.4 TB 2.4 GHz 16GB Rack (1U) Intel Xeon Silver 500 W DDR4-SDRAM (jw.com.au)	
Router	NETGEAR AX1800 Dual Band 4 Stream Wi-Fi 6 Router	Wireless networking across the campus with 1.8Gbps speeds, quad-core processor, advanced cyber threat protection. Link: NETGEAR AX1800 Dual Band 4 Stream Wi-Fi 6 Router - JB Hi-Fi (jbhifi.com.au)	\$209
Access Point	Ubiquiti (4 * \$209.29)	The Ubiquiti Unifi UAP-AC-LR is a long-range, ceiling-mounted wireless access point designed for extensive coverage, including a POE Injector for easy installation. It's ideal for educational environments, offering robust connectivity and scalability for Startech College's networking needs. Link: Ubiquiti Unifi UAP-AC-LR - Ceiling Mounted Wireless Access Point - 4Cabling	\$837.16
Switch	Cisco Business CBS250-48P-4X Smart Switch 48 Port GE PoE 4x10G SFP+ (\$2,369.86*2)	The Cisco CBS250 48-port GE Smart Switch offers advanced features for efficient network management. With 48 PoE ports and a 370W power budget, it supports high-power devices while ensuring energy efficiency, thanks to IEEE802.3az compliance. Security is enhanced through IEEE 802.1X port security and DoS attack prevention. Its intuitive interface simplifies operations, and its compact design is suited for various environments. The switch comes with a limited-lifetime warranty, ensuring reliability and peace of mind. Link: Cisco CBS250 48-port GE Layer 2 Smart Switch with max 370W PoE & 4x10G SFP+ (4cabling.com.au)	\$4739.72
Firewall	Fortinet FortiGate 60F	High throughput, VPN, integrated security, SD-WAN, easy management. Link: Fortinet FortiGate 60F Hardware – Next-Gen Firewall Protection & Security: Routers: Amazon.com.au	\$707.98
Patch Panels 2 * \$228.23	4Cabling 1RU 48 Port CAT6	T-568A/B wiring, RJ45 FCC compliant, for 22-26 AWG wires. Link: 4Cabling 1RU 48 Port Cat 6 Universal Termination Patch Panel - 4Cabling	\$456.46

<u>Network Cables</u> Display Port (\$41.18 * 2)	5m DisplayPort v1.4 Cable Male to Male 8K @60Hz Black	The director and receptionist require the 5m DisplayPort v1.4 Cable Male to Male in black, designed for superior video transmission to their external monitors. Link: 5m DisplayPort v1.4 Cable Male to Male. 8K @60Hz. Black - 4Cabling	
Ethernet Cables:	CAT6 LSZH LAN Ethernet Cable 305m - Reel Box: Blue (\$2,129.60*8 reels)	Student Computers: 60 units requiring 30 meters each, totalling 1,800 meters. Staff Computers: 8 units requiring 30 meters each, totalling 240 meters. Servers, Access Points, Cameras, Printers, Projectors: 19 units with varying cable length requirements, total length N/A. Server Room Computer: 1 unit requiring 30 meters, totalling 30 meters.	\$2211.96
Patch Cables:		Patch Cables Requirement: Total Number of Patch Cables: 94 Each Patch Cable Length: 2m Total Patch Cable Length: 188m Overall, the Ethernet infrastructure requires 2,040 meters of cable to connect all devices directly to the network room. Including the patch cables, the total cable length needed amounts to 2,228 meters. Note: CAT6 LSZH LAN Ethernet Cable pricing is based on 305m reels, with each reel priced at \$266.20. Link: Search results for: 'cat6 roll' (4cabling.com.au)	
Uninterruptible Power Supply (UPS)	APC Easy UPS On-Line SRV RM 2000 VA 230V with Rail Kit SRV2KRIRK	A double-conversion online UPS offers essential power protection and includes a bonus 1-year warranty, extending the total warranty coverage to 3 years. Link: APC Easy UPS On-Line SRV RM 2000 VA 230V with Rail Kit SRV2KRIRK (4cabling.com.au)	\$1,660.41
Network Tools and Testing Equipment	Wire stripper (2*\$8.36) Crimping tools (2 * \$120.12)	Essential tools for installation and testing.	\$322.86

	Network tester (2*\$32.95)	Link: Cable Continuity Tester 4Cabling	
Consultation Fee	Specialized IT Solutions Provider	IT infrastructure planning and setup consultation	\$3000

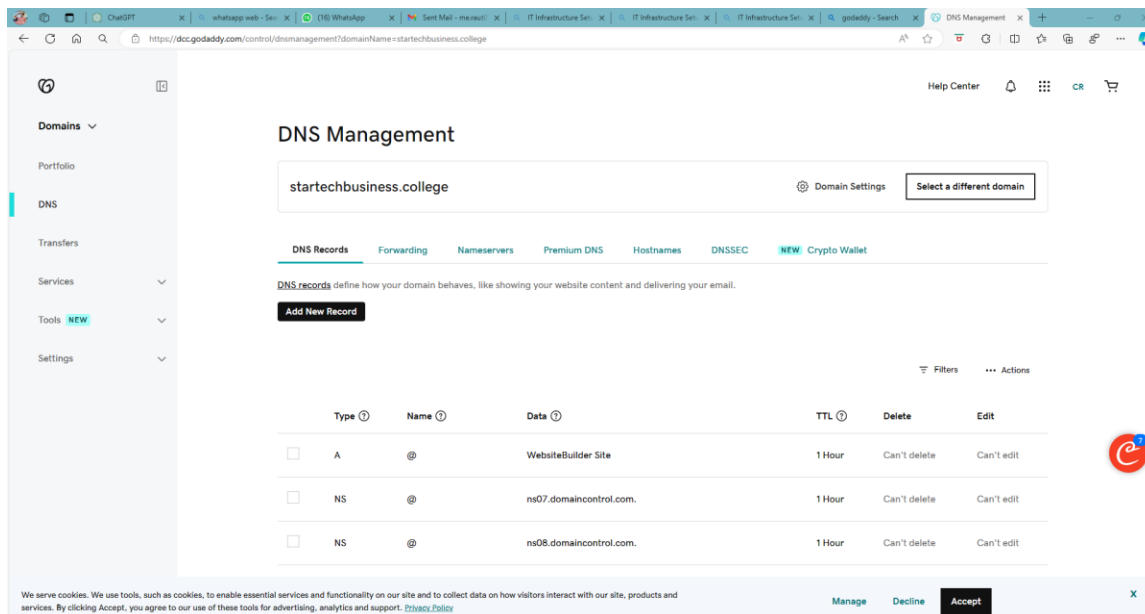
The estimated total cost for implementing the IT infrastructure setup plan at StarTech Business College, encompassing all required hardware, software, and services, is approximately AUD 123,369.35.

6. MICROSOFT 365 CLOUD INTEGRATION

The Microsoft 365 environment has been configured to provide centralized identity management, email services, and enterprise collaboration tools.

6.1 DOMAIN INTEGRATION

The domain startechbusiness.college has been successfully integrated with Microsoft 365 and Windows-based systems for authentication and directory services.



Microsoft 365 admin center

Home > Domains

Domains

+ Add domain Buy domain Refresh Filter Search domains

Domain name	Status
☐ startechbusiness.college (Default)	Healthy
☐ Chetanraut07.onmicrosoft.com	Healthy
☐ rauts.online	Healthy

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Domain Portfolio

startechbusiness.college

Overview DNS Products

DNS Records Forwarding Nameservers Premium DNS Hostnames DNSSEC

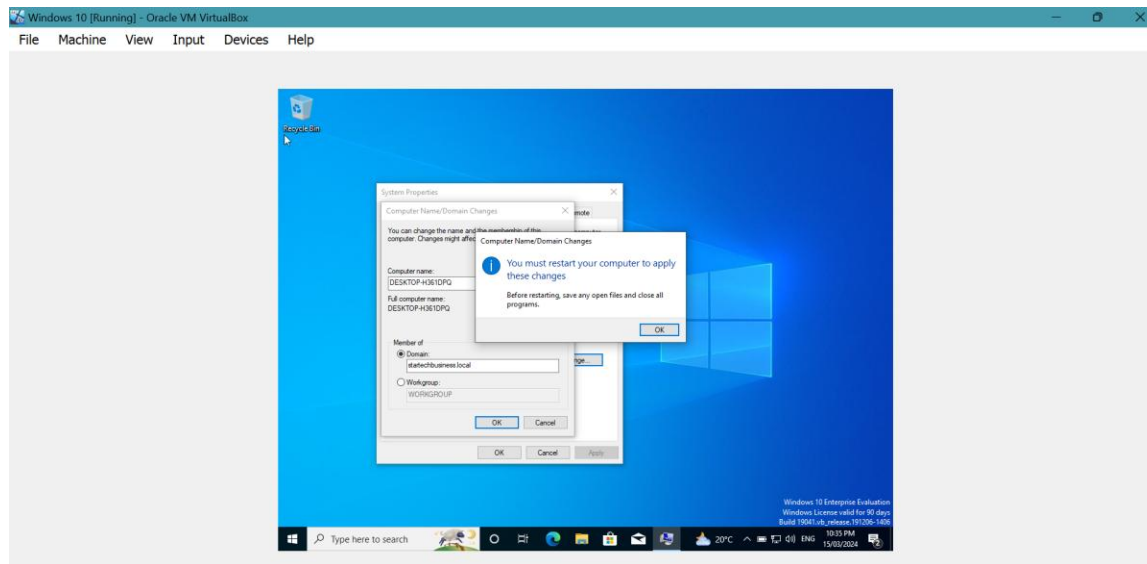
DNS records define how your domain behaves, like showing your website content and delivering your email.

New Records

TXT records are used to verify domain ownership, SSL verification, and email sender policies.

Type	Name	Value	TTL
TXT	startechbusiness.college	v=spf1 include:spf.protection.outlook.com ~all	1 Hour

Add More Records Save Cancel



Domain: startechbusiness.local

6.2 USER & ROLE MANAGEMENT

ADMINISTRATION TEAM

- Mehak Singh – Helpdesk Administrator
- Robin Maharjan – Helpdesk Administrator

ACADEMIC STAFF

- Amit Sharma
- Karan Patel

OPERATIONAL STAFF

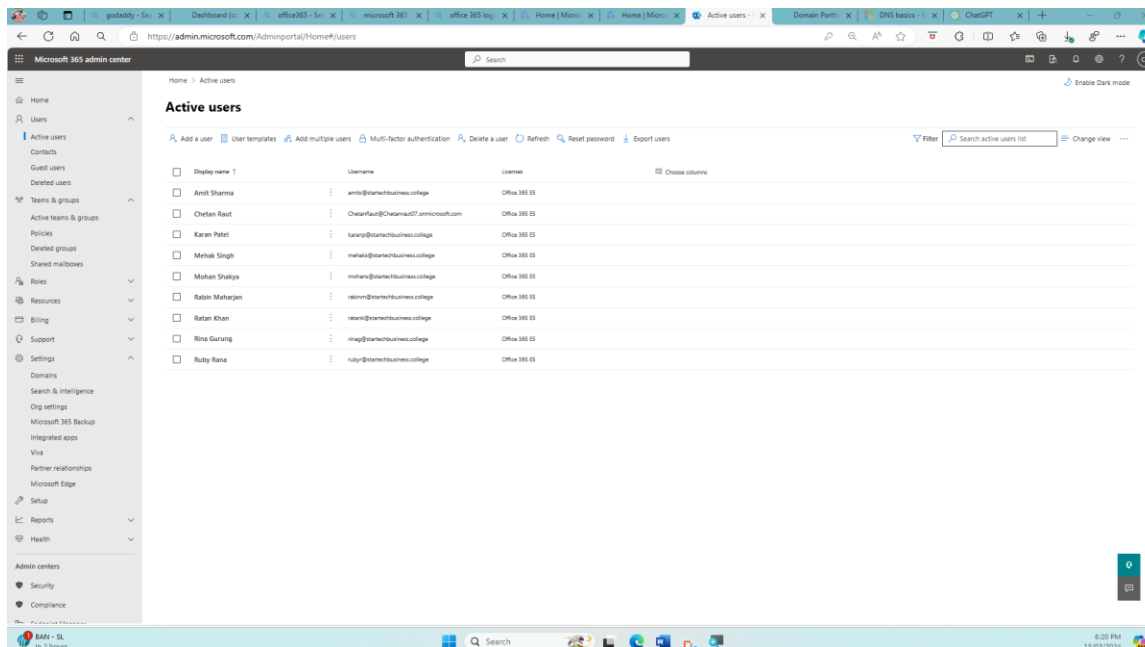
- Rina Gurung – Reception
- Mohan Shakya – Marketing
- Ruby Rana – Accounting

EXECUTIVE MANAGEMENT

- Ratan Khan – Global Administrator

All users are assigned Microsoft 365 licenses for full productivity access.

The screenshots below show the Microsoft 365 account created for the user.



7. IDENTITY & ACCESS MANAGEMENT

An Active Directory–based organizational structure has been implemented to support secure and controlled access across the environment.

7.1 ORGANIZATIONAL UNITS

Admin Team OU

- ❖ mehaks@startechbusiness.local
- ❖ rabinm@startechbusiness.local

Teachers OU

- ❖ amits@startechbusiness.local
- ❖ karanp@startechbusiness.local

Marketing OU

- ❖ mohans@startechbusiness.local

Accounting OU

- ❖ rubyr@startechbusiness.local

Management OU

- ❖ ratank@startechbusiness.local

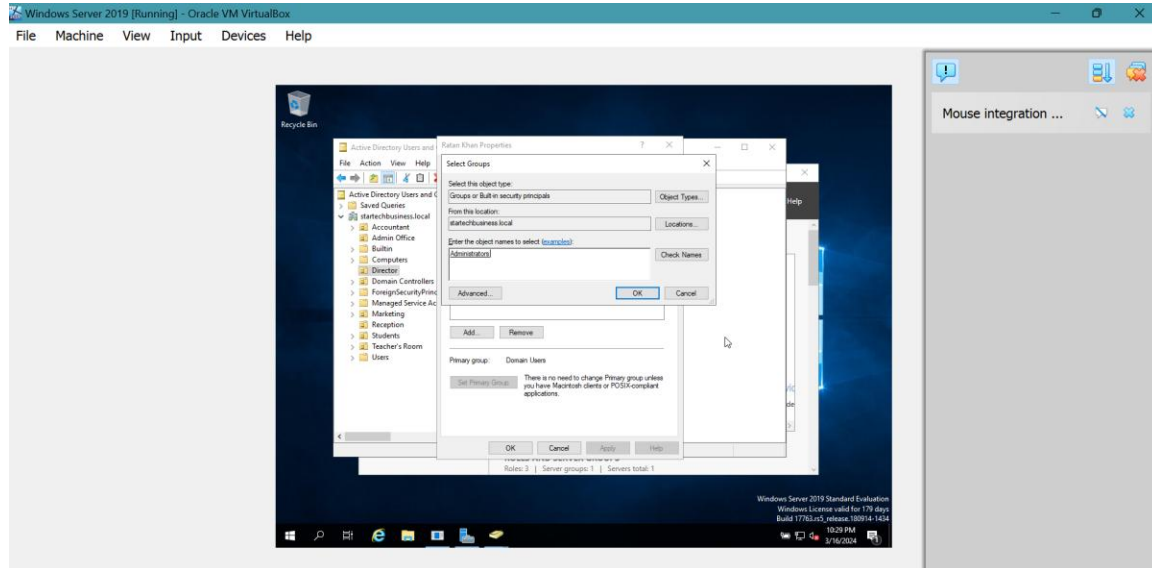
Reception OU

- ❖ rinag@startechbusiness.local

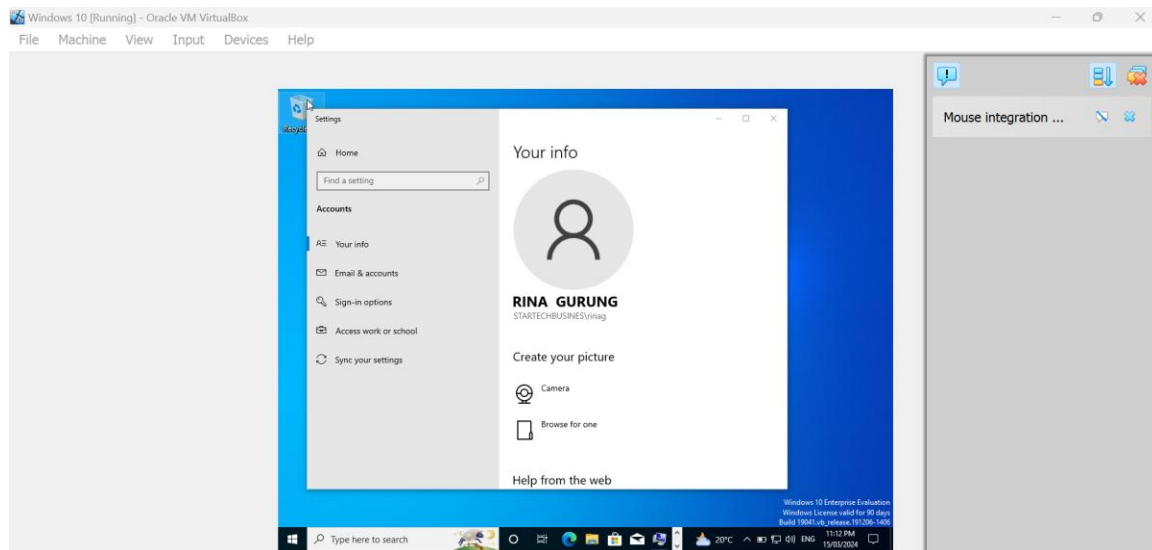
7.2 ACCESS CONTROL MODEL

- Role-based access control enforced across departments
- Administrative privileges restricted to designated roles
- Cross-department access applied where required
- Centralized control maintained through domain administration

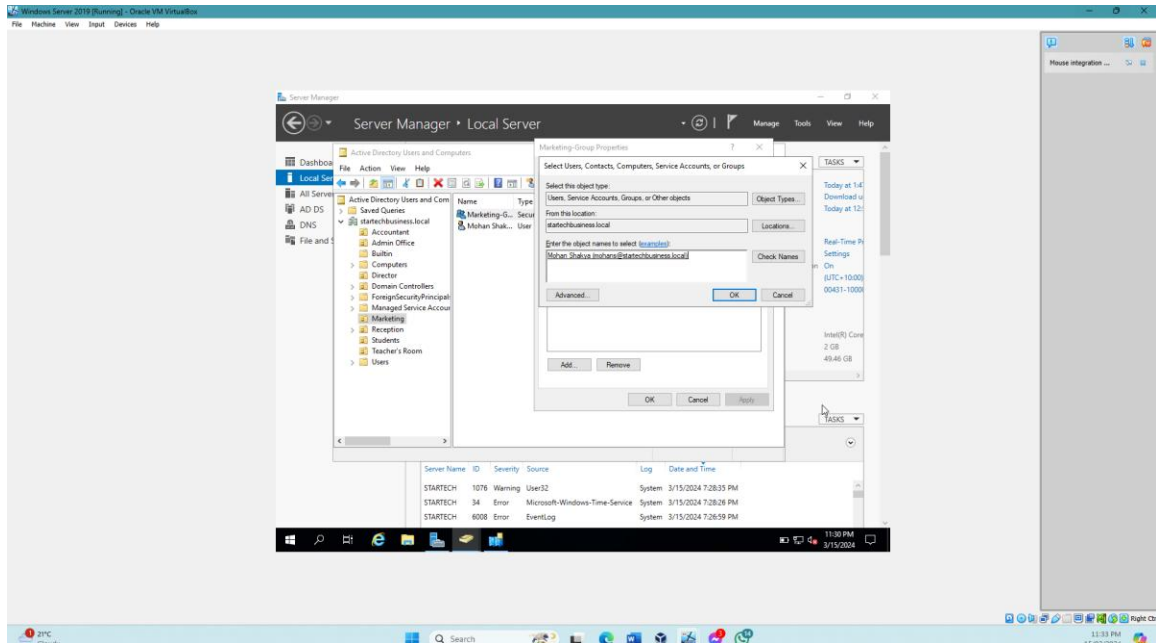
Administrative privileges were assigned to the Director role to enable centralized control and system management.



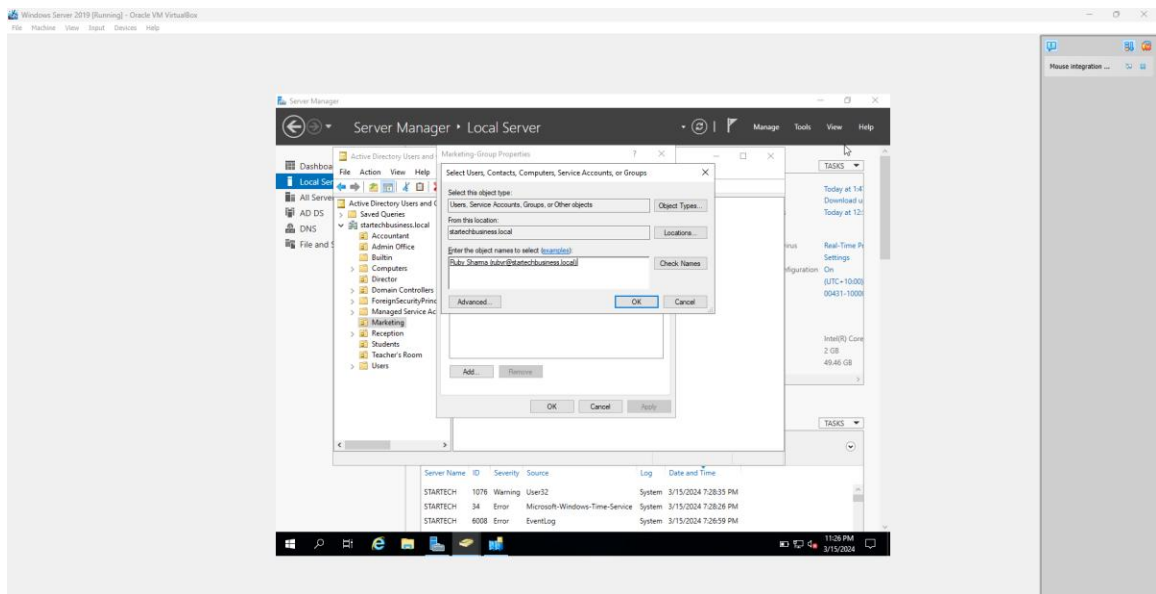
Rina Gurung has successfully logged into Windows 10.



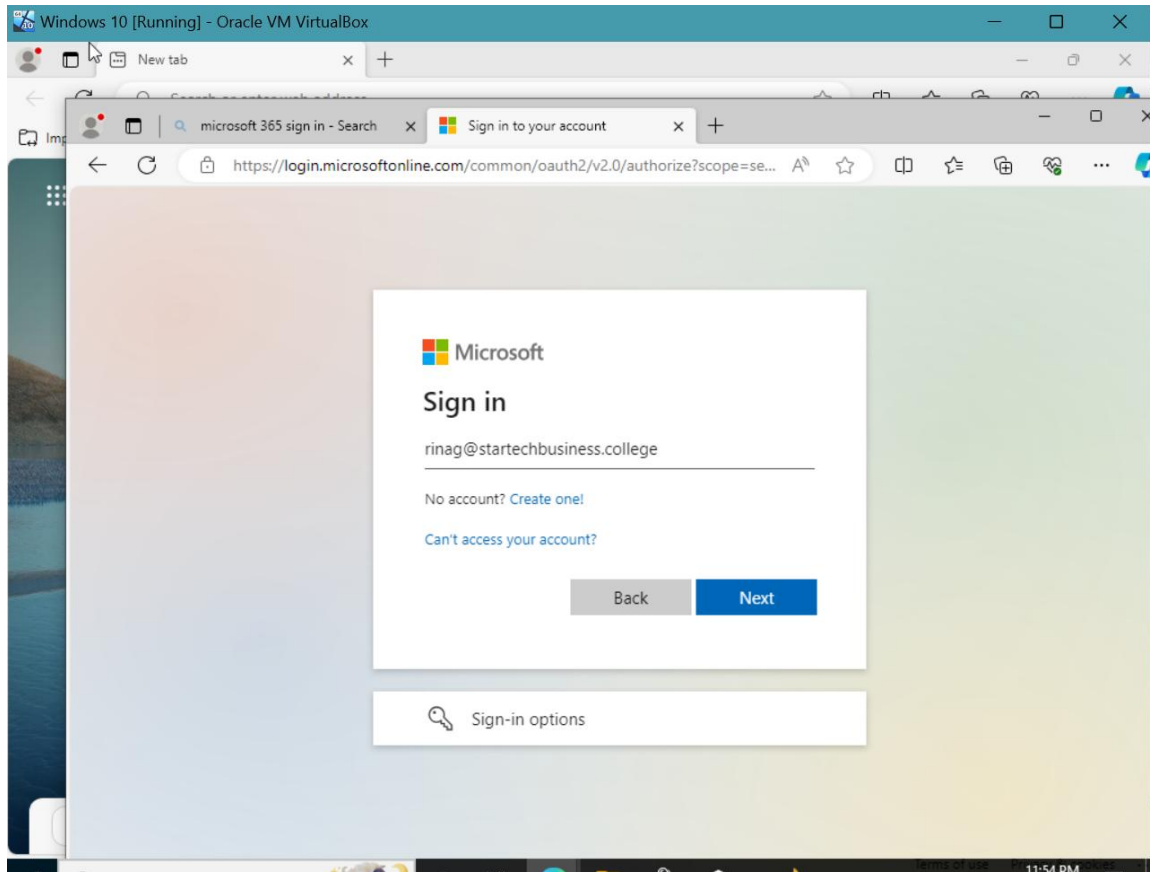
Highlighting how users are being added to their respective security groups.

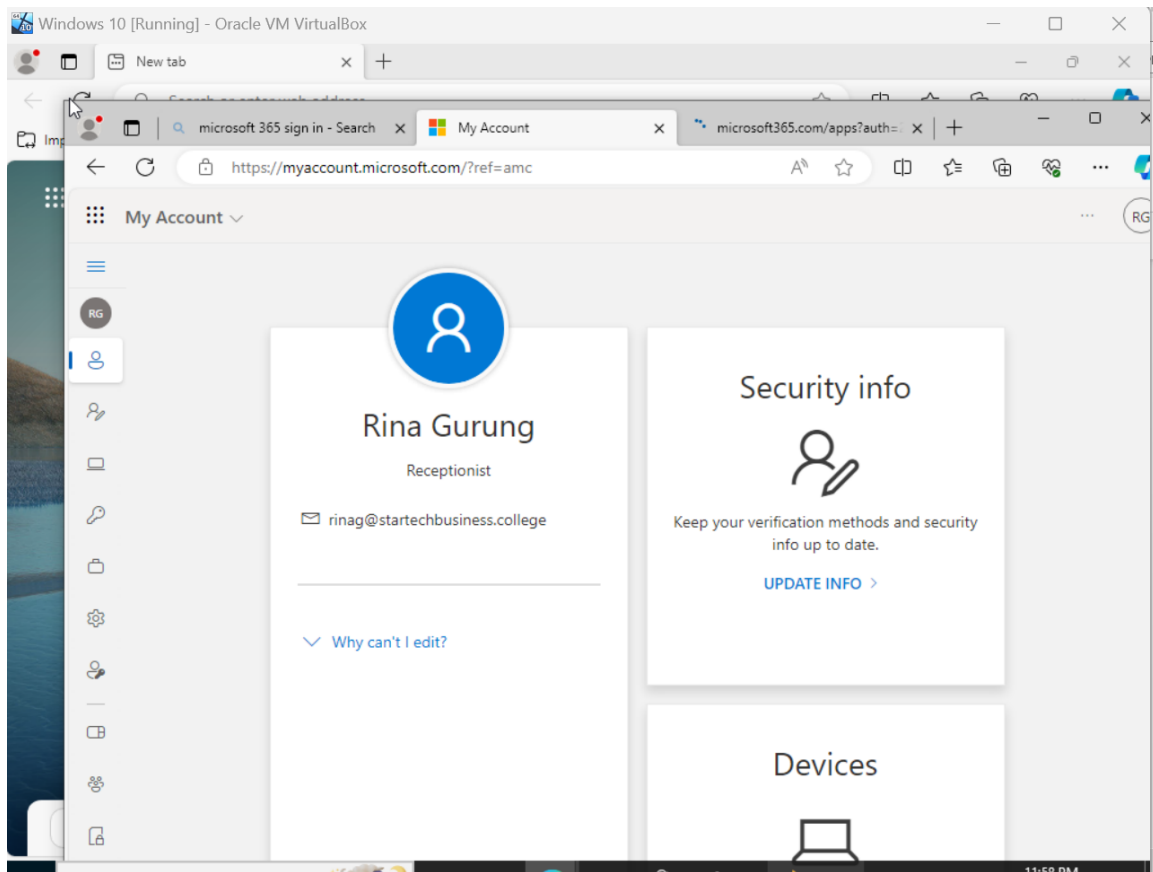


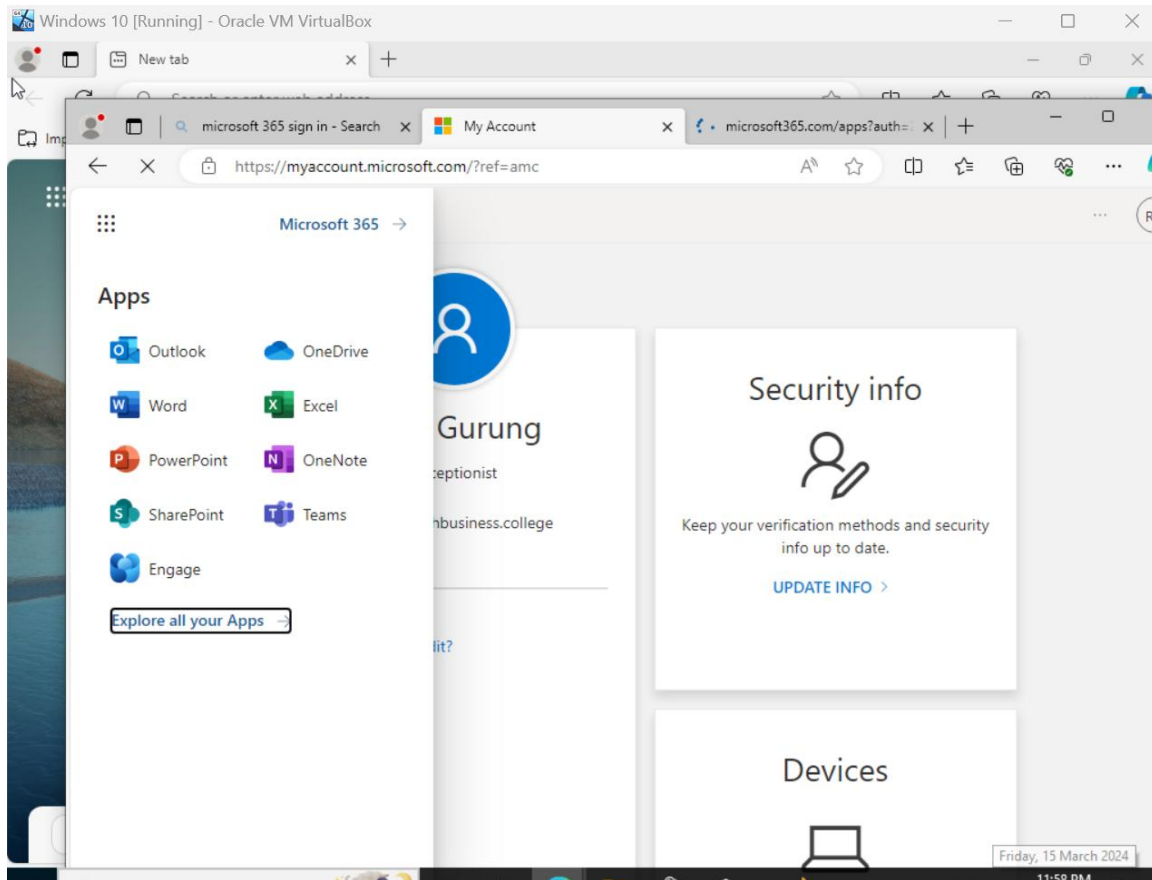
Ruby has also been granted access to the marketing department.



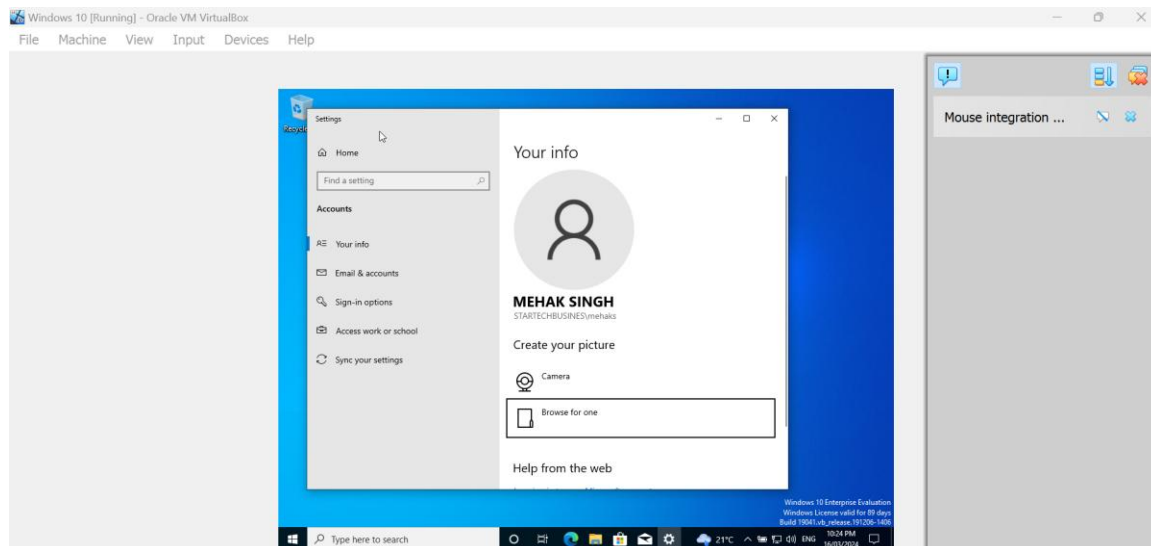
Rina Gurung is logging into the Microsoft 365 account assigned to her, enabling access to Office 365 licenses.







Screenshot highlighting the next user from Admin Office name **Mehak Singh** successfully logged into windows 10.



8. SYSTEM VALIDATION

The infrastructure has been validated across key operational areas:

- Domain integration successfully completed
- User authentication and login verified
- Microsoft 365 services fully operational
- Network connectivity established across all systems
- Security policies and access controls enforced successfully

9. CONCLUSION

The StarTech Business College IT infrastructure design delivers a complete enterprise-grade technology environment combining networking, systems infrastructure, cloud services, and security architecture into a unified operational framework.

The solution ensures scalability, security, and centralized management across all departments while supporting both academic and administrative functions. The integration of on-premises infrastructure with Microsoft 365 cloud services enables modern identity management and operational efficiency.

This design reflects real-world enterprise IT architecture principles, including structured networking, secure identity management, infrastructure scalability, and cost-efficient deployment planning.